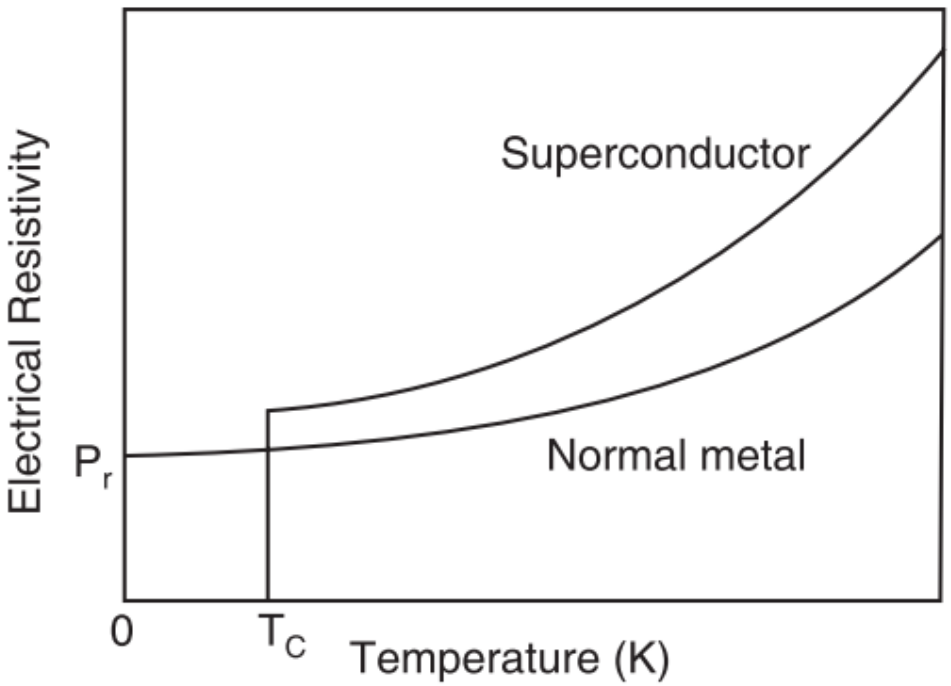
GITAM University

Subject-Physics

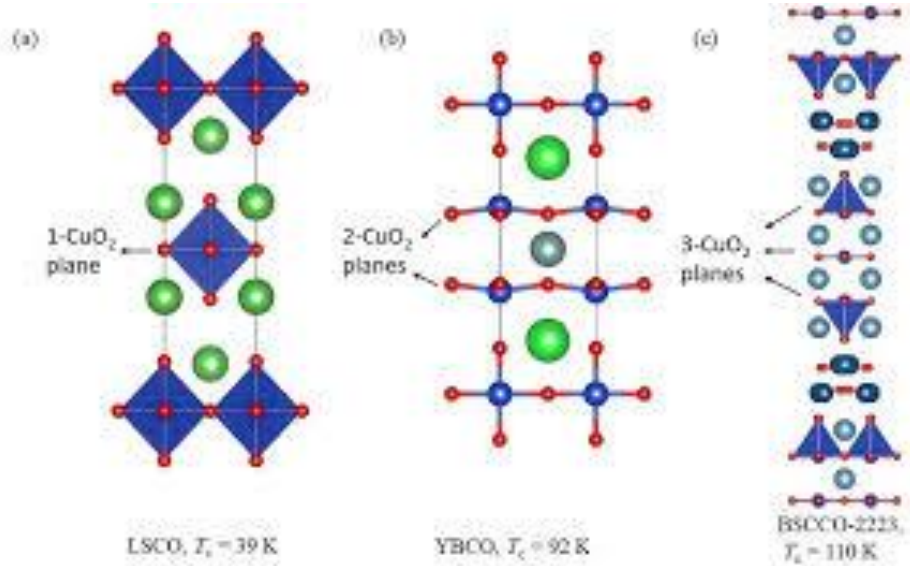
Topic: Superconducting material

Sayan Chaudhuri


Electrical Resistivity
Superconductor
Normal metal
P_r
0
T_C
Temperature (K)

H																	He
Li	Be 0.03											B	C	N	O	F	Ne
Na	Mg											Al 1.19	Si 6.7	P 4.6-6.1	S	Cl	Ar
K	Ca	Se	Ti 0.39	V 5.3	Cr	Mn	Fe	Co	Ni	Cu	Zn 0.9	Ga 1.09	Ge 5.4	As 0.5	Se 6.9	Br	Kr
Rb	Sr	Y 0.5;2.7	Zr 0.55	Nb 9.2	Mo 0.92	Tc 7.8	Ru 0.5	Rh 325 μ	Pb	Ag	Cd 0.55	In 3.4	Sn 3.7: 5.3	Sb 3.6	Te 4.5	I	Xe
Ca 1.5	Ba 1.8: 5.1	La 4.8;5.9	Hf	Ta 4.4	W 0.01	Re 1.7	Os 0.65	Ir 0.14	Pt	Au	Hg 4.15 3.95	Ti 2.39 1.45	Pb 7.2	Bi 3.9 7.2;8.5	Po	At	Rn
Fr	Ra	Ac															
			Ce 1.7	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu 0.1-0.7	
			Th 1.37	Pa 1.3	U 0.2	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lw	

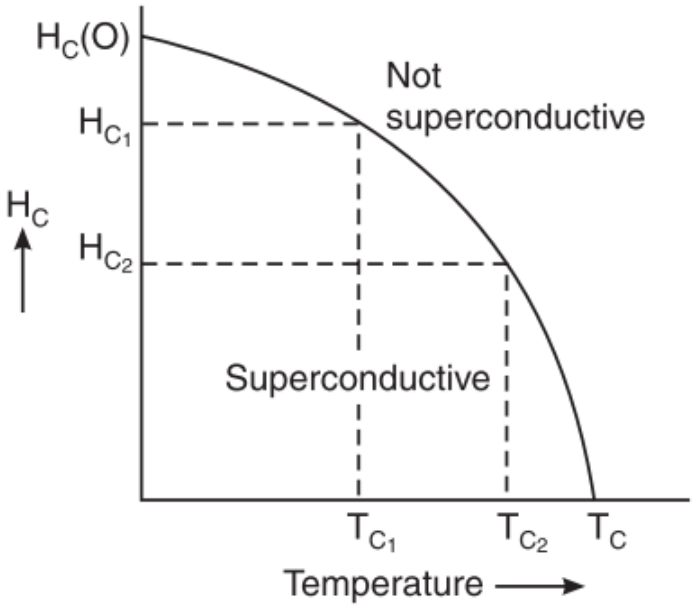
High T_C superconductors

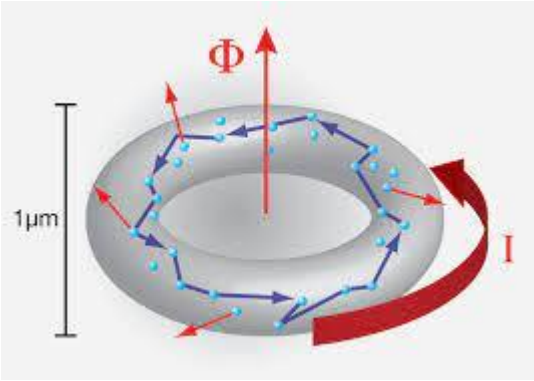


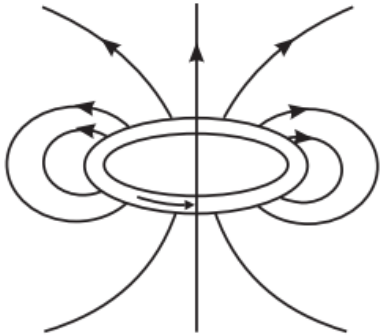
compound	T_C(K)
$Nd_{1.85}Ce_{0.15}CuO_4$	24
$La_{1.85}Sr_{0.15}CuO_4$	40
$YBa_2Cu_3O_7$	92
$Bi_2Sr_2Ca_2Cu_3O_{10}$	110
$Tl_2Ba_2Ca_2Cu_3O_{10}$	127
$Hg_2Ba_2Ca_2Cu_3O_8$	134

PROPERTIES OF SUPERCONDUCTORS

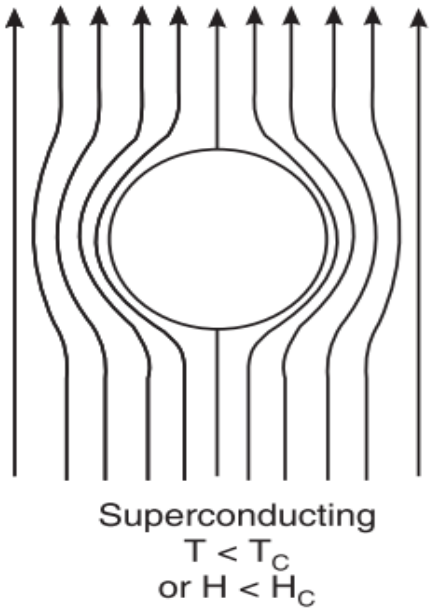
- **No electrical Resistance**
- **Persistent Current** can be generated
- Have fixed **Critical Temperature, T_C.**
- Have certain **Critical Magnetic Field, H_C**
- **Critical Current Density:** Silsbee effect







Meissner Effect



Superconductor becomes a perfect diamagnetic below T_C.

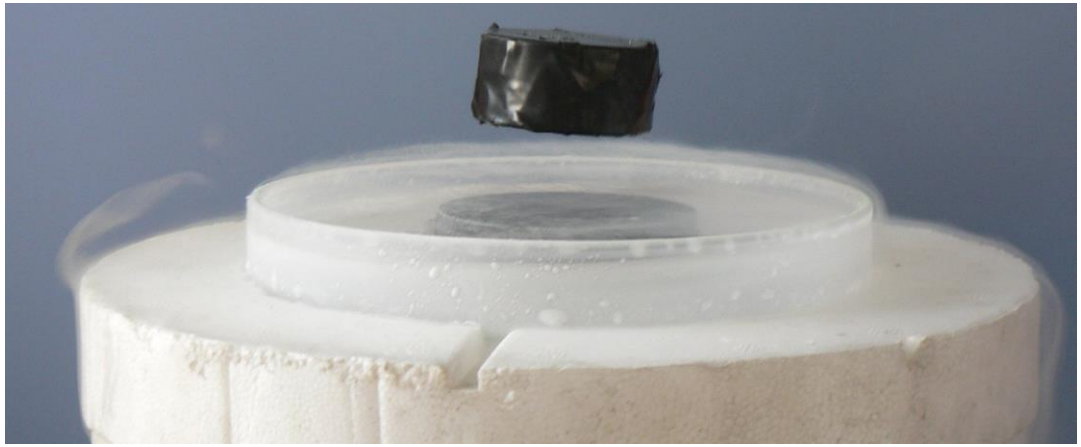
$$\chi = \frac{M}{H} = -1$$

London Penetration Depth

$$H(x) = H(0)e^{-\frac{x}{\lambda}}$$

λ ranges between 300 Å to 5000 Å

$$\lambda(T) = \frac{\lambda(0)}{\sqrt{\left[1-\left(\frac{T}{T_C}\right)^4\right]}}$$



Type of Superconductors

Type-I Superconductors

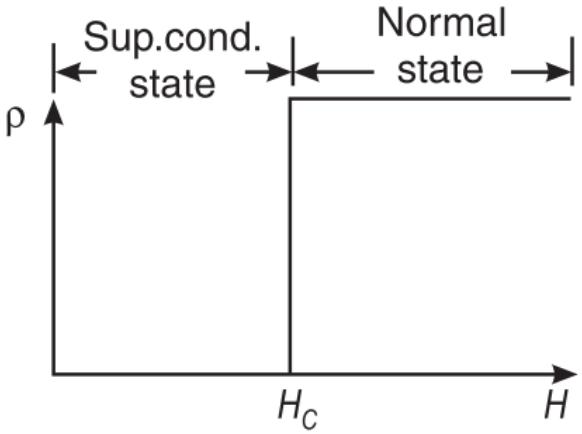
Exm: Aluminium, lead and indium

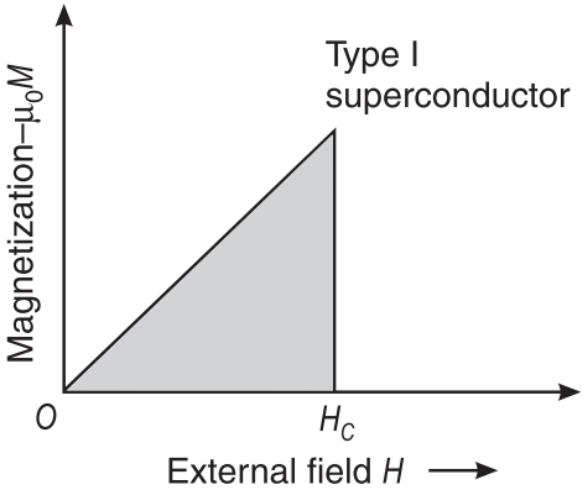
1. They are perfectly diamagnetic and exhibit complete Meissner effect.

2. They have only one critical field. At the critical field the magnetization drops to zero.

3. The maximum critical field for type I superconductor is of the order of 0.1 Wb/m^2.

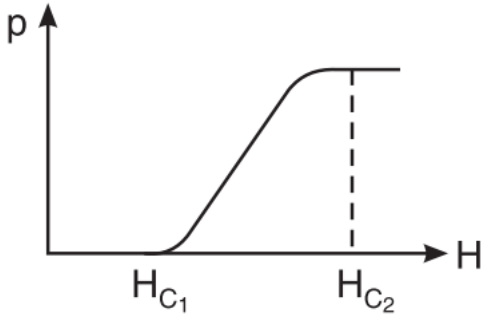
4. The transition at H_C is reversible. Below HC the material behaves as a superconductor, and above H_C it behaves as a normal conductor.

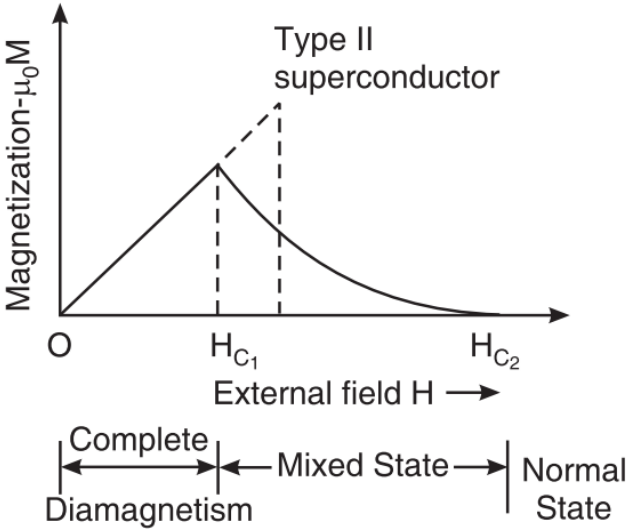


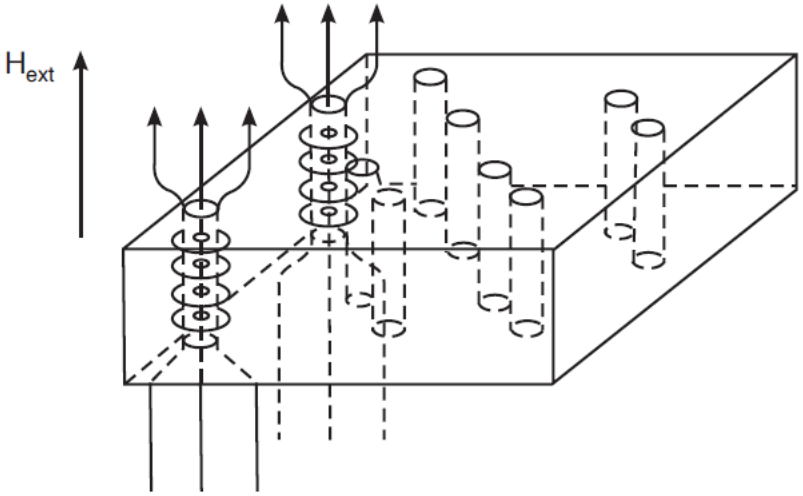


Type-II Superconductors

1. They have two critical magnetic fields
2. The material is perfect diamagnetic below the lower critical field, H_{C1}.
3. Above H_{C1} they do not show complete Meissner effect
4. They exist in an intermediate state in between the critical fields, H_{C1} and H_{C2}.
5. At H_{C2} the magnetization vanishes and the specimen returns to normal conducting state.
6. The upper critical field is very high and is of the order of 30 Wb/m^2.







Sl.No.	Type-I Superconductors	Type-II Superconductors
1.	They exhibit complete Meissner effect	They do not exhibit complete Meissner effect
2.	They show perfect diamagnetic behaviour	They do not show perfect diamagnetic behaviour
3.	They have only one critical magnetic field, H_C	They have two critical magnetic fields, lower critical magnetic field, H_{C1} and upper critical magnetic field, H_{C2}
4.	There is no mixed state or intermediate state in case of these materials	Mixed state or intermediate state is present in these materials
5.	The material loses magnetization abruptly	The material loses magnetization gradually
6.	Highest value for H_C is about 0.1 Wb/m^2	Upper critical field is of the order of 30 Wb/m^2
7.	They are known as soft superconductors	They are known as hard superconductors
8.	Lead, tin, mercury are examples	Nb-Sn, Nb-Ti, Nb-Zr, Va-Ga are examples

Isotope Effect

Critical temperature decreases with increasing isotopic mass M

$$\boldsymbol{T_C \propto M^{-\alpha} \quad \therefore M^{\alpha} T_C = Constant}$$

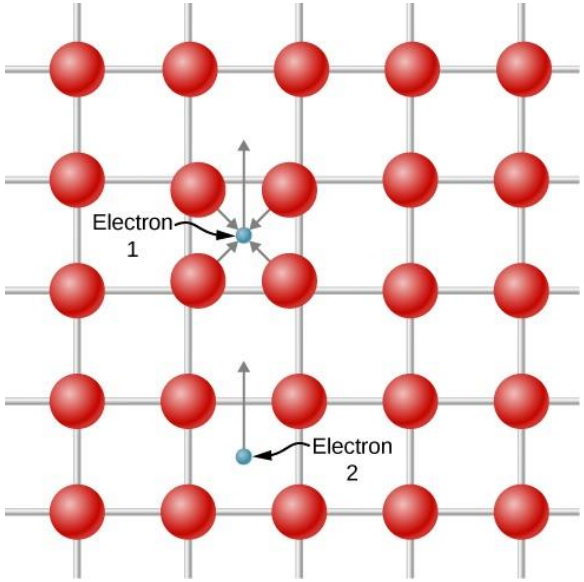
Change in the isotopic mass does not change the electronic structure. Dependence of the transition temperature on the isotopic mass suggests involvement of electron-lattice interaction.

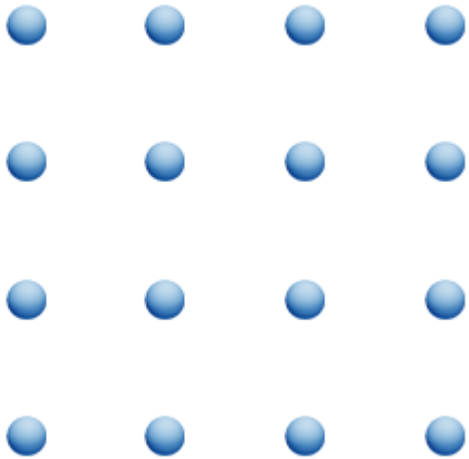
BCS THEORY (Bardeen, Cooper, Schrieffer)

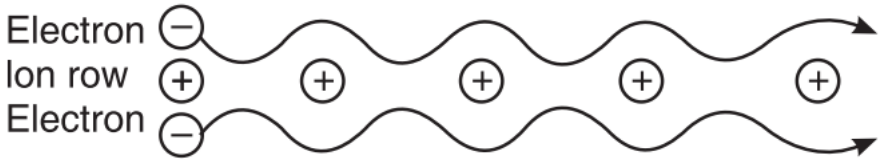
- Electrons form pairs, called Cooper pairs, which propagate throughout the lattice.
- Such propagation is without resistance because the electrons move in resonance with phonons.

$$E_g(0) = 2\Delta \cong 3.52kT_C$$

$E_g \sim 10^{-3}$ eV


Electron
1
Electron
2

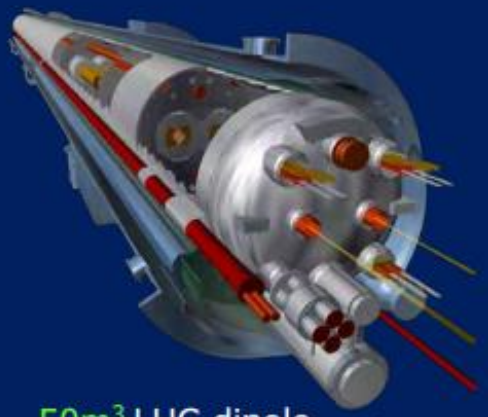



Electron
Ion row
Electron

0.0001 m^3 HF insert model
~ **200 A**

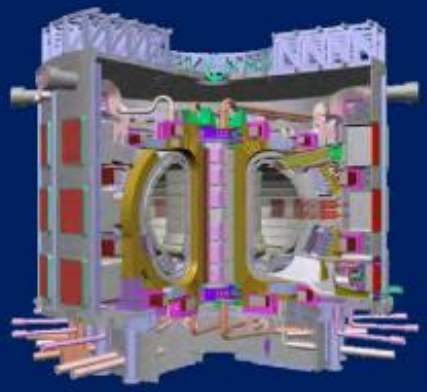
2 m^3 MRI magnet
200-800 A @ 1-3 T, ~10 MJ

25 m^3 ATLAS solenoid
8 kA @ 2T, 40 MJ



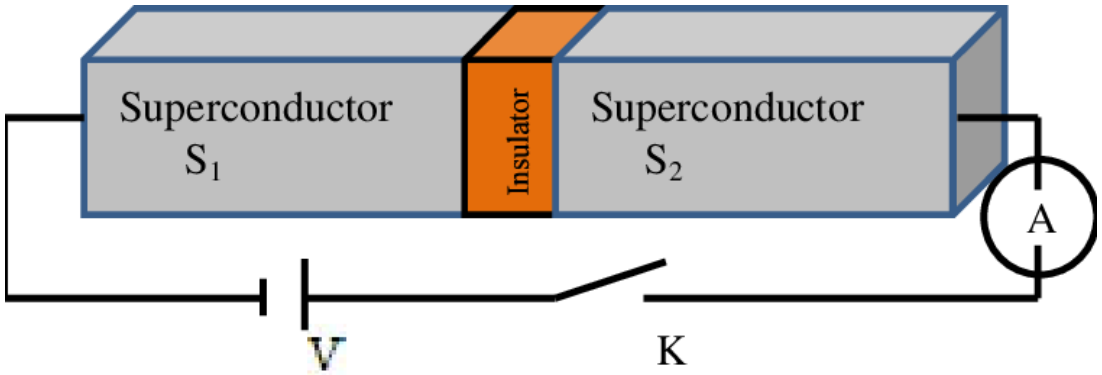
50m^3 LHC dipole
12 kA @ 8.3 T

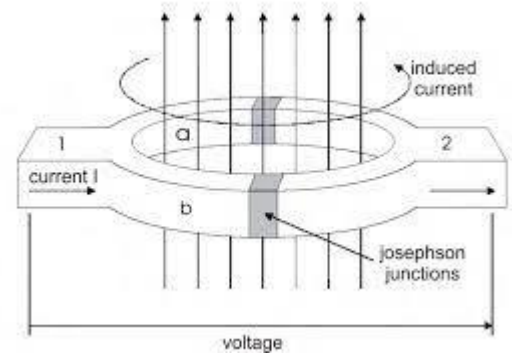
400 m^3 HEF detector magnet
20 kA @ 4 T, 2.6 GJ



1000 m^3 ITER magnets
40-70 kA @ 10-13T , 50 GJ

JOSEPHSON EFFECT





dc Josephson Effect

$$I_S = I_C Sin\phi_0$$

ϕ_0 is phase diff between two SCs

ac Josephson Effect

$$I_S = I_C Sin(\phi_0 + \Delta\phi)$$

$\Delta\phi$ is phase diff due to DC field

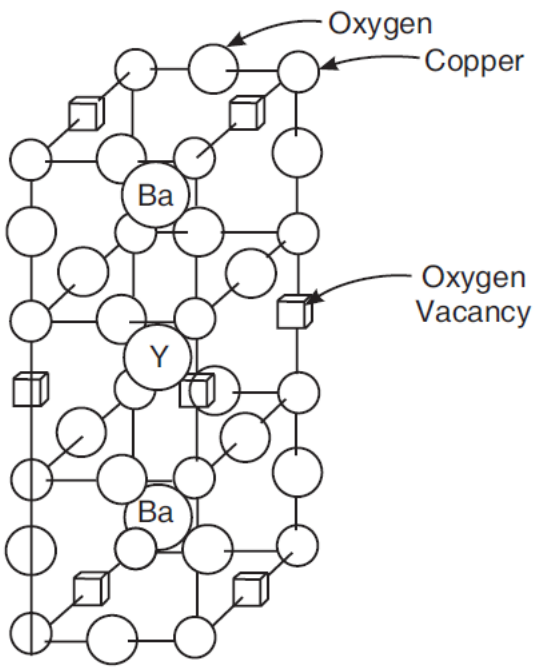
$$\Delta\phi = 2\pi t\left(\frac{2eV}{h}\right)$$

Generate ac current with $\omega = \frac{2eV}{h}$

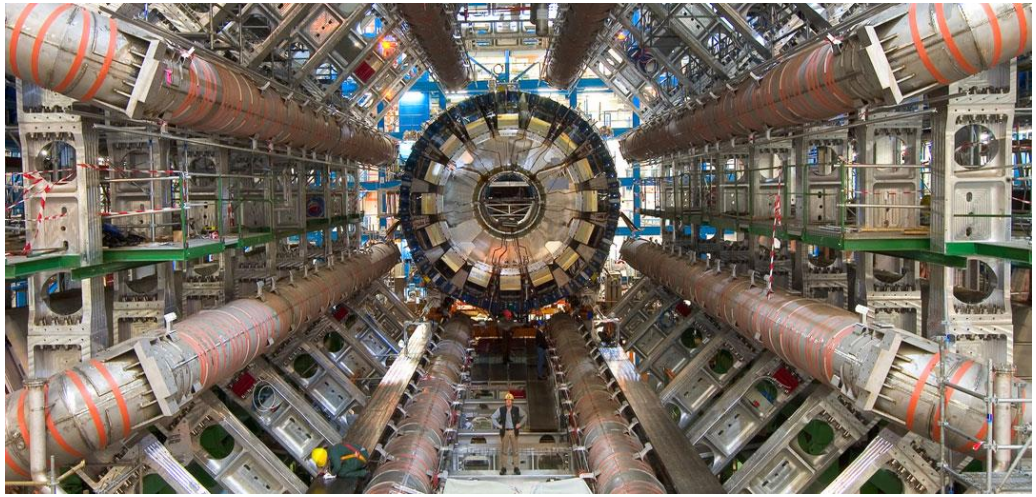
High T_C Superconductors

T_C > 27 K

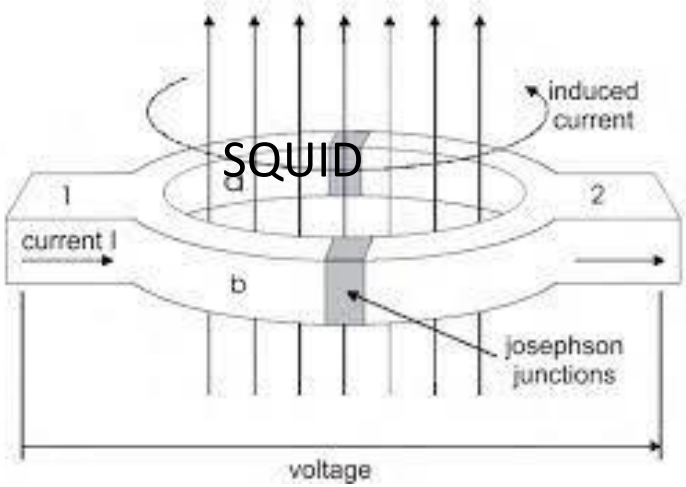
	T_C
$Y\,Ba_2\,Cu_3O_7$	92
$Bi_2Sr_2Ca_2Cu_3O_{10}+\delta$	105
$Th_2Ca\,Ba_2\,Cu_2O_{10}+\delta$	125



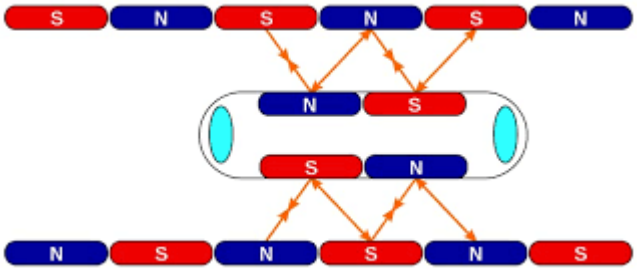
Superconducting magnets:

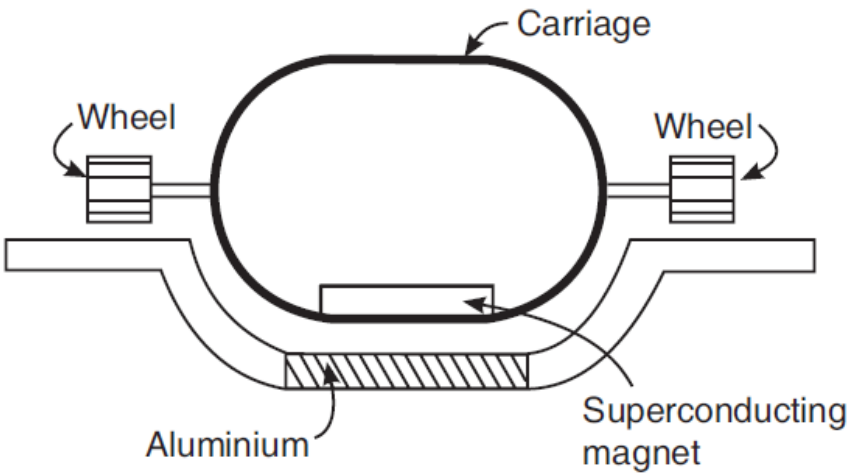


SQUID



MAGLEV train





Problems

1. The transition temperature for Pb is 7.2 K. However, at 5 K it loses the superconducting property if subjected to magnetic field of 3.3×10^4 A/m. Find the maximum value of H which will allow the metal to retain its superconductivity at 0 K.

2. The critical field of niobium is 1×10^5A/m at 8 K and 2×10^5A/m at 0 K. Calculate the transition temperature of the element.

3. Calculate the penetration depth of lead at 5.2 K if the London penetration depth at 0 K is 37 nm. The critical temperature of lead is 7.193 K.